

Level	What it Studies	NLP Focus	Example
Phonology	Sounds of language (phonemes) and sound rules	Speech recognition, text-to-speech, pronunciation modeling	*“cats” → /kæts/ vs. “dogs” → /dɔ:gz/ (plural sounds differ)
Morphology	Structure of words, morphemes (roots, prefixes, suffixes)	Stemming, lemmatization, morphological segmentation	“played” → <i>play</i> + <i>ed</i> ; “unbelievable” → <i>un</i> + <i>believe</i> + <i>able</i>
Syntax	Sentence structure, grammar, word order	Parsing, grammar checking, machine translation	“ <i>The man eats chicken</i> ” vs. “ <i>The chicken eats man</i> ” (same words, different syntax = different meaning)
Semantics	Literal meaning of words/sentences	Word embeddings, semantic similarity, question answering	“ <i>car</i> ” ≈ “ <i>automobile</i> ” ; “ <i>bank</i> ” (<i>money</i>) ≠ “ <i>bank</i> ” (<i>river</i>)
Pragmatics	Meaning in context , speaker intent, implied meaning	Dialogue systems, intent detection, sarcasm detection, discourse analysis	“ <i>Can you pass the salt?</i> ” → Not about ability, it’s a request “ <i>Wow, you’re such a great driver!</i> ” (after someone crashes) Semantics: compliment. Pragmatics: actually a sarcastic insult .